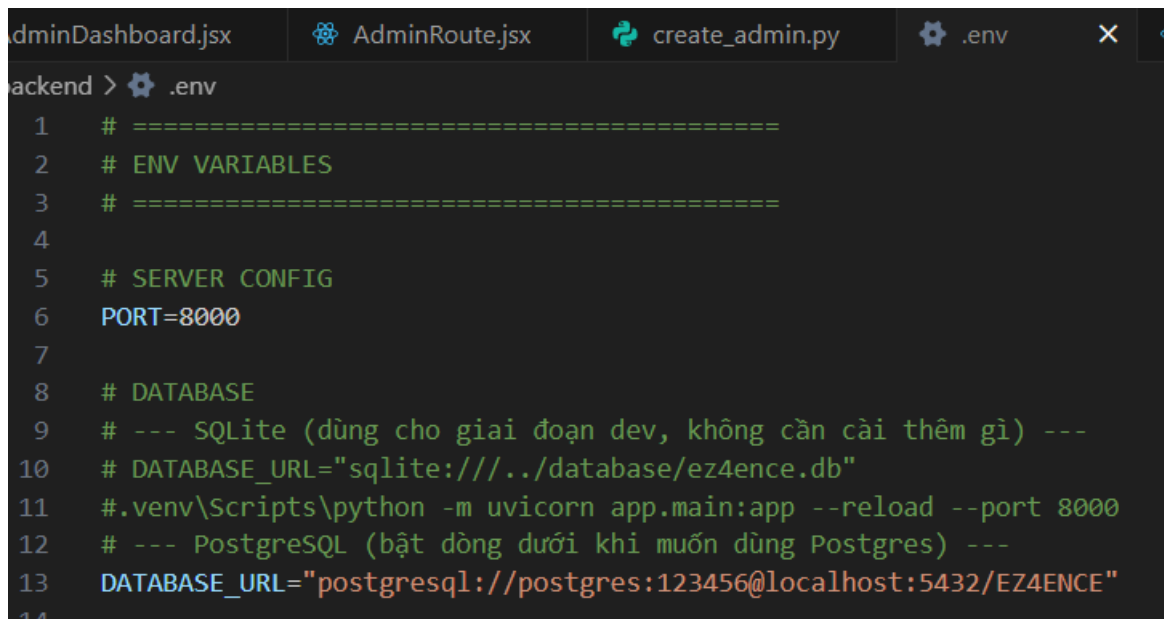
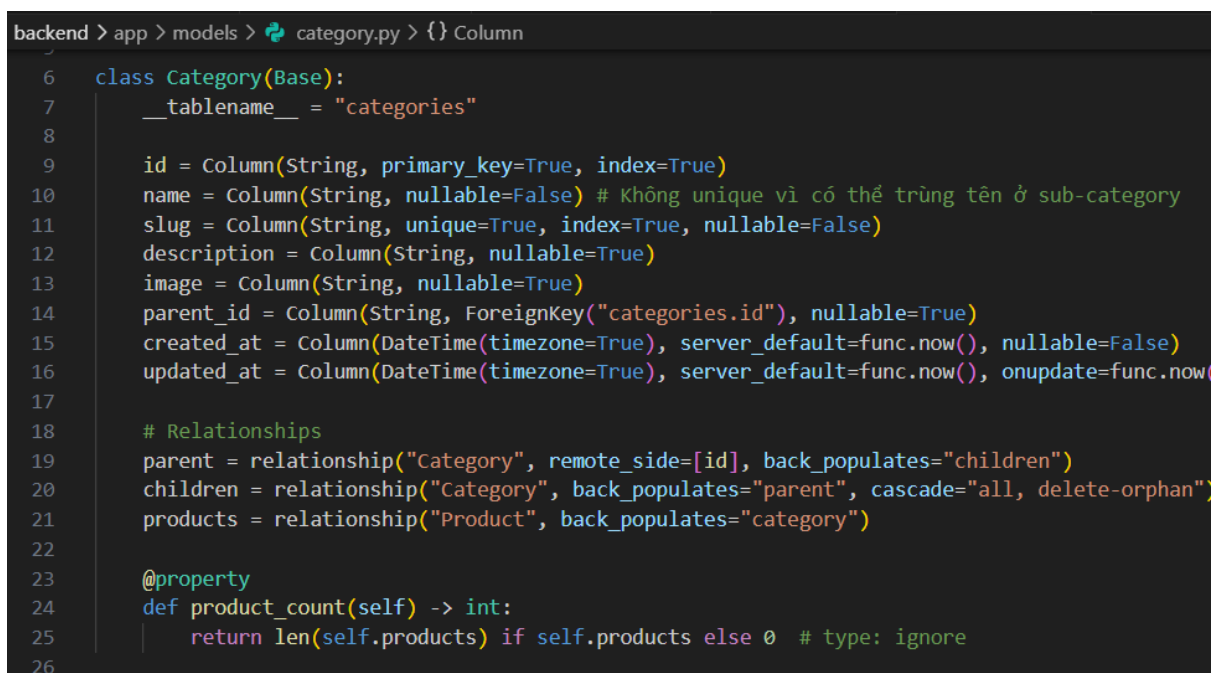


SESSION 8: PostgreSQL Integration



```
adminDashboard.jsx | AdminRoute.jsx | create_admin.py | .env | X
backend > .env
1 # =====
2 # ENV VARIABLES
3 # =====
4
5 # SERVER CONFIG
6 PORT=8000
7
8 # DATABASE
9 # --- SQLite (dùng cho giai đoạn dev, không cần cài thêm gì) ---
10 # DATABASE_URL="sqlite:///../database/ez4ence.db"
11 #.venv\Scripts\python -m uvicorn app.main:app --reload --port 8000
12 # --- PostgreSQL (bật dòng dưới khi muốn dùng Postgres) ---
13 DATABASE_URL="postgresql://postgres:123456@localhost:5432/EZ4ENCE"
14
```

Hình 1: kết nối với postgresQL



```
backend > app > models > category.py > {} Column
5
6 class Category(Base):
7     __tablename__ = "categories"
8
9     id = Column(String, primary_key=True, index=True)
10     name = Column(String, nullable=False) # Không unique vì có thể trùng tên ở sub-category
11     slug = Column(String, unique=True, index=True, nullable=False)
12     description = Column(String, nullable=True)
13     image = Column(String, nullable=True)
14     parent_id = Column(String, ForeignKey("categories.id"), nullable=True)
15     created_at = Column(DateTime(timezone=True), server_default=func.now(), nullable=False)
16     updated_at = Column(DateTime(timezone=True), server_default=func.now(), onupdate=func.now())
17
18     # Relationships
19     parent = relationship("Category", remote_side=[id], back_populates="children")
20     children = relationship("Category", back_populates="parent", cascade="all, delete-orphan")
21     products = relationship("Product", back_populates="category")
22
23     @property
24     def product_count(self) -> int:
25         return len(self.products) if self.products else 0 # type: ignore
26
```

Hình 2: Bảng category có các khóa chính, khóa ngoại lấy từ DB

The screenshot shows a PostgreSQL database interface. On the left, the Object Explorer displays a tree view of the database schema, including tables like 'addresses', 'banners', 'brands', 'cart_items', 'carts', 'categories', 'chat_messages', 'chat_sessions', 'compatibility_overrides', 'news', 'order_items', 'order_status_history', 'orders', 'product_images', 'product_skus', 'products', 'reviews', 'sku_images', 'stock_receipt_items', and 'stock_receipts'. The 'products' table is selected.

The Query editor at the top shows the following SQL query:

```
SELECT * FROM public.products
ORDER BY id ASC
```

The Data Output pane at the bottom displays the query results in a table format. The table has three columns: 'id', 'name', and 'slug'. The results show a list of products, including items like 'Loa JBL Quantum Duo Gaming 2.0 RGB', 'Loa Edifier R12800Bs Bluetooth Active Bookshelf 2.0', 'VGA Sapphire PULSE AMD Radeon RX 7900 GRE 16GB', and 'Laptop Gaming HP OMEN 16-wd0013TX'.

Hình 3: dữ liệu thật trong DB